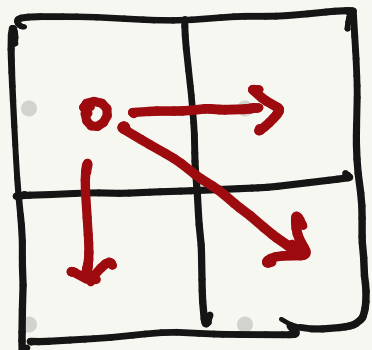


Homework-7

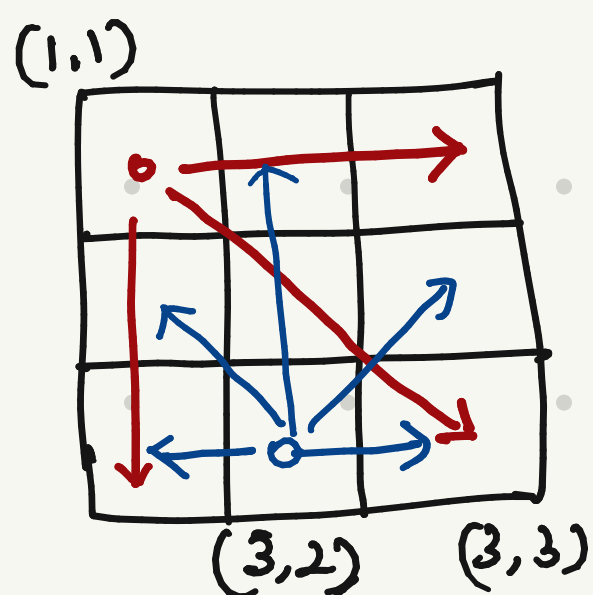
- ② Instances where  $n$ -queens has no solution. There are only two values of  $n$  for which  $n$ -queens has no solution.

→  $n = 2$



Any two places on this board either share a row, a column or lie on the same diagonal. Therefore two queens always attack each other.

→  $n = 3$



In this board after placing the queen at  $(1,1)$  we have only two remaining positions to place the second queen  $\{(3,2), (2,3)\}$ . If we pick any one of these positions, all the remaining positions are in attack by these two queens. Hence, a third queen cannot be placed on the board and there is no solution.

- ③  $W = 52$   
 $w_1 = 2, w_2 = 10, w_3 = 13, w_4 = 17, w_5 = 22, w_6 = 42$

From algorithm 5.4,

initial values:  $i = 0, \text{weight} = 0, \text{total} = 2 + 10 + 13 + 17 + 22 + 42 = 106$

• Path(13, 17, 22)

→ Step-1

$i = 1, w_1 = 2, \text{weight} = 0, \text{total} = 104$

include[1] = no

promising  $(0 + 104 > 52 \text{ and } 0 + 10 <= 52) \rightarrow \text{Yes}$

→ step-2

$i = 2, w_2 = 10, \text{weight} = 0, \text{total} = 94$

include[2] = no

promising  $(0 + 94 > 52 \text{ and } 0 + 13 <= 52) \rightarrow \text{Yes}$

→ Step-3

$$i=3, w_3=13, \text{weight}=13, \text{total}=81$$

$$\text{include}[3] = \text{yes}$$

$$\text{promising}(13+81 > 52 \text{ and } 13+17 \leq 52) \rightarrow \text{Yes}$$

→ Step-4

$$i=4, w_4=17, \text{weight}=30, \text{total}=64$$

$$\text{include}[4] = \text{yes}$$

$$\text{promising}(30+64 > 52 \text{ and } 30+22 \leq 52) \rightarrow \text{Yes}$$

→ Step-5

$$i=5, w_5=22, \text{weight}=52, \text{total}=42$$

$$\text{include}[5] = \text{yes}$$

$$\text{weight} = 52 \rightarrow \text{goal reached}$$

$$\text{subset} = \{13, 17, 22\}$$

• Path(10, 42)

→ Step-1

$$i=1, w_1=2, \text{weight}=0, \text{total}=104$$

$$\text{include}[1] = \text{no}$$

$$\text{promising}(0+104 > 52 \text{ and } 0+10 \leq 52) \rightarrow \text{Yes}$$

→ Step-2

$$i=2, w_2=10, \text{weight}=10, \text{total}=94$$

$$\text{include}[2] = \text{yes}$$

$$\text{promising}(10+94 > 52 \text{ and } 10+13 \leq 52) \rightarrow \text{Yes}$$

→ Step-3

$$i=3, w_3=13, \text{weight}=10, \text{total}=81$$

$$\text{include}[3] = \text{no}$$

$$\text{promising}(10+81 > 52 \text{ and } 10+17 \leq 52) \rightarrow \text{Yes}$$



→ Step-4

$$i=4, w_4=17, \text{weight}=10, \text{total}=64$$

include[4] = no

promising ( $10+64 \geq 52$  and  $10+22 \leq 52$ ) → Yes

→ Step-5

$$i=5, w_5=22, \text{weight}=10, \text{total}=42$$

include[5] = no

promising ( $10+42 \geq 52$  and  $10+42 \leq 52$ ) → Yes

→ Step-6

$$i=6, w_6=42, \text{weight}=52, \text{total}=0$$

include[6] = yes

weight == 52 → goal reached

subset = {10, 42}

Solutions = { {10, 42}, {13, 17, 22} }